

WEEK 39 INTRO TO POPULATION GENETICS

BOOK EXERCISES



DEPARTMENT OF MOLECULAR BIOLOGY AND GENETICS

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EVOLUTIONARY THINKING
29. AUGUST 2025

EMMA FEMKE DIEPEVEEN
PH.D.-STUDERENDE



PRIMATE OF THE WEEK

Siamang Gibbon

Symphalangus syndactylus

Largest gibbon → part of the lesser apes

Found in south-east Asia

Have a “Gular sac” → inflates before and during vocalisations → can inflate to twice the size of the siamangs head → very loud (as loud as a chainsaw)

They have longer arms than their legs and their big toe is opposable too

They have no natural predators (but are still endangered :()

Dads are very involved in the offspring management <3

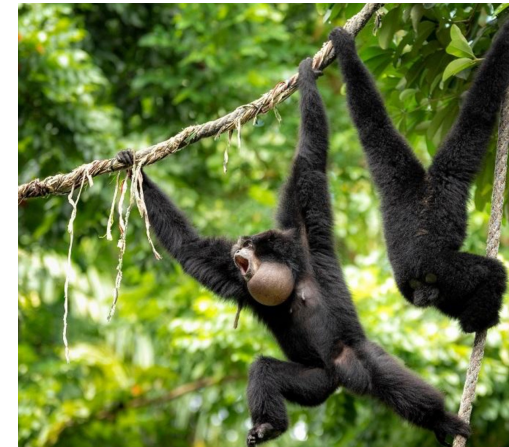


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PLAN FOR TODAY

1. Little recap of the main concepts

2. In your own group

Make the book exercises of chapter 1,2 & 3



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CHAPTER 2: GENETIC DRIFT & MUTATION

- **Genetic drift**: random changes in allele frequencies in populations of finite size
- **Mutation**: new alleles arise at rate μ per generation.
- **Wright-Fisher model**:
 - Population of $2N$ gene copies
 - Each new copy picks a parent at random from the previous generation
 - Independent, equal-probability sampling (like drawing marbels from a bag)
 - When no mutations: every allele is eventually fixed or lost



DRIFT

→ in **smaller** populations, by chance you can just have a shift in allele frequency by chance → **more effect of drift**.

This means a **temporary crash in N erases genetic variation** fast → you can see this a population bottleneck or when you have a founder effect.



IF A **NEW NEUTRAL MUTATION** APPEARS IN A POPULATION OF $N=5000$ DIPLOID INDIVIDUALS – WHAT IS THE PROBABILITY IT EVENTUALLY BECOMES **FIXED**?

- A. $1 / 5.000$
- B. $1 / 10.000$
- C. $5.000 / 10.000$
- D. it depends on μ



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 - D. it depends on μ

A new mutation is just 1 out of the total of $2N$ copies.

→ The $\text{pr}(\text{fixation}) = 1/(2N) = 1 / 10.000$



MUTATION AND MOLECULAR CLOCK

1. Fixation probability

$$\Pr(\text{fixation}) = f_A$$

Any single gene copy is **equally** likely to be the eventual ancestor of everyone → so an allele's chance of **fixing equals its current frequency**

2. Substitution rate

$$\text{Rate} = 2N\mu * 1/(2N) = \mu$$

More mutations arise in bigger populations, but each is proportionally less likely to fix → the 2N cancels out → also means that substitution rate is independent of population size!!

3. Molecular clock

$$E[d] = 2\mu t$$

Differences between 2 species **accumulate at a constant rate** in both species, over time



CHAPTER 3: COALESCENCE THEORY

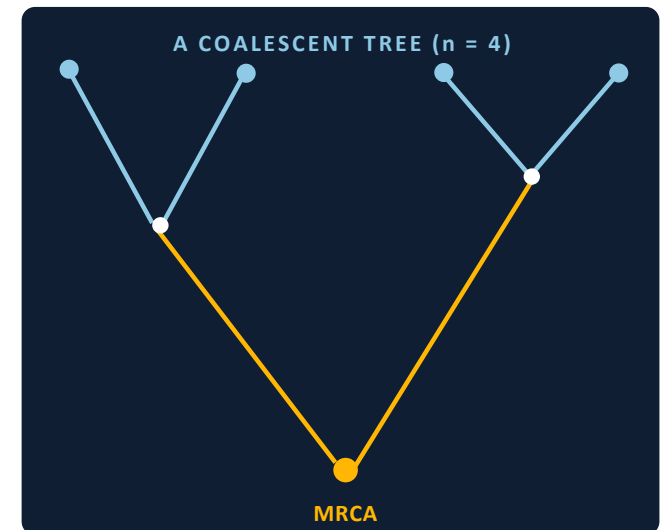
Same idea but then **back in time**

Forward in time (chapter 2): start with a population, ask how **allele frequency** change going forward

Backward in time (chapter 3): start with a sample of DNA sequences, **trace their ancestry** back until lineages merge (aka coalesce) at a common ancestor

Pr (2 lineages coalesce after 1 generation) = $1/(2N)$

→ Each of the $2N$ gene copies in the previous generation is equally likely to be its parent



$\theta = 4Nm$ → LINKS DIVERSITY TO POPULATION SIZE

$$\theta = 4N\mu$$

2 lineages * 2N generations (mean coalescence time) * mutation rate μ

θ = expected number of nucleotide differences between 2 sequences

Bigger N → deeper coalescence times → more time for mutations → more diversity

Effective population size N_e → the size of an idealized Wright-Fisher population that would show the same amount of drift as the real one

(N_e is different than the total population because of fluctuating population size, unequal sex ratios, and offspring variance)



AFRICANS HUMAN POPULATIONS SHOW **HIGHER θ** THAN NON-AFRICAN POPULATIONS, EVEN THOUGH POPULATION SIZES CAN BE **FAR SMALLER**. WHAT IS THE BEST EXPLANATION?

A) Higher mutation rate in Africa

B) Larger effective population size (N_e)

C) More sequencing errors

D) Stronger selection outside Africa



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Africans did not have have a bottle neck → non-African populations had a out-of-Africa bottleneck → lost a lot of diversity → larger N_e in africans → thus larger θ → because $4N_e\mu$



TODAY

Exercises from the book

Chapter 1 → questions 1.1, 1.2, 1.3, 1.4

Chapter 2 → questions 2.1, 2.2, 2.3, 2.4, 2.5

Chapter 3 → questions 3.1, 3.2, 3.3, 3.4

Answers are in the back of the book and online on the github.

Make them with your group!

Tip: what are they asking and find which equation answers that ?



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FRIDAY – PYTHON EXERCISES

You will work on [Wright-fisher](#) and [Coalescent simulations](#) exercises in python



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